

# Ketan Sharma

Data Science & AI Engineer · Computer Vision · LLMs & Agentic Systems

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## PROFESSIONAL SUMMARY

Data Science & AI engineer (MSc, UAL Creative Computing Institute, London) specialising in computer vision, deep learning, and LLM-powered systems. Previously Crowd Technical Director at DNEG, building large-scale simulations for film and television with Python and C++. Hands-on with the full lifecycle of AI products — data pipelines, model training, RAG and agentic workflows, and deployment of end-to-end systems with web dashboards and automation. Reliable team player with strong communication skills, seeking roles in AI engineering, machine learning, and data science.

## CORE SKILLS

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|--------------------|----------------------------------------------------------------------------------------------------------|
| Programming        | Python, C++, JavaScript, SQL                                                                             |
| Machine Learning   | PyTorch, TensorFlow, Keras, Scikit-learn, model training & evaluation, data augmentation                 |
| AI Specialisations | Computer Vision, Deep Learning, NLP, Generative AI, LLM integration, RAG, agentic AI, prompt engineering |
| Data & Tooling     | Pandas, NumPy, Matplotlib, SQLite, data wrangling & analysis, web scraping (BeautifulSoup, Selenium)     |
| Web & Systems      | Flask, REST APIs, real-time systems, multi-threading, automation pipelines, Git                          |
| Cloud & MLOps      | AWS, Azure, Google Cloud Platform, local LLM deployment (Ollama), scalable deployments                   |

## FEATURED PROJECTS

### Business Radar — Automated Lead Generation & AI Outreach Platform

End-to-end system that finds local businesses, analyses their web presence with AI, and runs a complete personalised email outreach pipeline — built solo in Python.

Python

Flask

SQLite

Local LLMs (Ollama)

Web Scraping

Selenium

SMTP / IMAP

Automation

- Multi-source business discovery:** modular scrapers for Google Maps/Places, Yell, Yelp, and FreeIndex behind a common scanner engine, with deduplication keys, lead-quality validation, and rate limiting.

- **AI-personalised outreach at zero API cost:** locally hosted LLMs analyse each business's website (with Selenium screenshot capture) and generate tailored cold emails, then automate the full lifecycle — SMTP sending with safety limits, day-3/7/12 follow-ups, and IMAP reply tracking with classification.
- **Lead-management dashboard:** Flask web app with a pipeline view, AI draft review, reply inbox, interactive lead map, and analytics on an SQLite persistence layer — production-minded with dry-run mode and a rate-limited send queue.

Code: [github.com/ketans7/business-radar](https://github.com/ketans7/business-radar) · Live demo: [ketans7.github.io/demo/business-radar/](https://ketans7.github.io/demo/business-radar/)

## RiderBro — AI Decision Assistant for Delivery Riders

Telegram bot that turns a screenshot of a Deliveroo order offer into an instant accept / borderline / skip verdict, grounded in the real cycling route — plus a React operator dashboard over the collected data.

Python

FastAPI

React

Telegram Bot

OCR + Local VLM

Mapbox Routing

SQLite

Geospatial

- **Multimodal screenshot understanding:** OCR with a local vision-language model fallback (MLX / Ollama) extracts the payout and every pickup/dropoff address fully on-device — screenshots never leave the rider's machine.
- **Real-time decision engine:** computes the actual cycling route via Mapbox through every stop, prices in per-stop and peak-hour friction, and returns a verdict with effective £/hour and a per-leg time breakdown in seconds.
- **Self-improving data flywheel:** every order, route, outcome, and post-delivery survey is logged to SQLite, powering positioning suggestions from OpenStreetMap venue data and a React analytics dashboard with live feed, spatial heatmaps, and hunting-ground maps.

Code: [github.com/ketans7/RiderBro](https://github.com/ketans7/RiderBro) · Live demo: [ketans7.github.io/demo/riderbro/](https://ketans7.github.io/demo/riderbro/)

## PROFESSIONAL EXPERIENCE

### Crowd Technical Director — DNEG

Jan 2022 – Jun 2023

Mumbai, India · Film & television VFX

- Developed and implemented large-scale crowd simulations for major film and television productions.
- Applied algorithmic problem-solving in Python and C++ to optimise crowd behaviours and improve the efficiency of complex, large-scale scenes.
- Collaborated with VFX teams to integrate simulations into shots, balancing technical accuracy with artistic direction.
- Managed technical workflows and asset structures across large projects, keeping pipelines reliable and accessible for the whole team.

## EDUCATION

### MSc Data Science & AI for the Creative Industries

2024 – 2025

UAL Creative Computing Institute, London, UK

- Specialised in computer vision, deep learning, and real-time AI applications.
- Focus on applying AI algorithms to creative and industrial problem-solving.

### BSc Animation, Interactive Technology, Video Graphics & Special Effects

2020 – 2023

Amity University Mumbai, India

- Strong technical foundation in creative technology, digital content creation, and technical workflows.

LANGUAGES & INTERESTS

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**Languages:** English (fluent), Hindi (native), Thethi (native), Marathi (conversational)

**Interests:** running and weightlifting, aquarium keeping, quantum computing, cooking.